

ROTUA IMMANUELA TAMPUBOLON

Information Systems Student · Software Development · System Analysis · QA · Product

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PROFESSIONAL SUMMARY

Versatile Information Systems student with hands-on experience across software development, system analysis, UI/UX design, quality assurance, and business process improvement. Proven ability to deliver end-to-end technical projects — from requirements elicitation and system documentation to implementation, testing, and deployment. Comfortable working across the full technology stack and project lifecycle, with strong technical literacy and structured documentation skills. Actively building a project portfolio that spans multiple domains, with a consistent focus on clean architecture, real-world problem solving, and professional-grade deliverables.

EDUCATION

Institut Teknologi Del — Situluama, Indonesia
Bachelor of Information Systems | 2024 – Present | GPA: **3.45 / 4.00**

TECHNICAL SKILLS

Programming	Java · Python · JavaScript (TypeScript) · C · SQL · HTML
Frameworks	Node.js · Express.js · Prisma ORM · JDBC · Maven
Databases	PostgreSQL · MySQL · ERD Design · Schema Normalization
System Analysis	Requirements Elicitation · SRS/SyRS · Use Case Narratives · DFD · BPMN 2.0 · Data Dictionary · Sequence Diagrams
Testing & CI/CD	Manual Testing · API Testing (Postman · Newman) · Test Case Design · Bug Reporting · GitHub Actions
Design & Collab	Figma · Draw.io · Bizagi Modeler · Git · GitHub · Notion
Soft Skills	Analytical Thinking · Technical Documentation · Cross-functional Collaboration · Problem Solving · Attention to Detail

PROJECTS

Game Inventory System — Java OOP CLI Application 2025

OOP Course Project · Team of 4 · github.com/RotuaTampubolon/game-inventory

- Built a CLI-based RPG Inventory Management System in Java applying multi-level inheritance (Item → Equipment → Weapon → MeleeWeapon), interfaces (Equippable, Usable, Tradeable), and polymorphism across 15+ classes
- Implemented JDBC with PostgreSQL for full CRUD using the Data Mapper pattern, separating persistence from business logic across mapper, repository, and service layers
- Integrated Singleton, Factory design patterns and custom exceptions (ItemNotFoundException, InventoryFullException) for robust error handling; applied Java Collection Framework (HashMap, PriorityQueue, Stack, Set) for inventory operations
- Collaborated using Git with feature branching — each member owned an end-to-end feature slice (model → mapper → service → CLI)

Tech Stack: Java 17 · PostgreSQL · JDBC · Maven · Jansi · Git

Studify — Student Productivity REST API 2025

Personal Project · github.com/RotuaTampubolon/studify

- Built a fullstack REST API using Node.js, Express, PostgreSQL, and Prisma ORM following layered architecture (routes → controllers → services → database)
- Implemented JWT-based authentication with bcrypt password hashing, stateless token middleware, and ownership-based authorization preventing unauthorized cross-user access
- Developed Task CRUD endpoints with status tracking (TODO / IN_PROGRESS / DONE), priority levels, and filtering — all routes protected by JWT verification

Tech Stack: Node.js · Express · PostgreSQL · Prisma ORM · JWT · bcrypt · Git

MindSpend — Behavioral Finance Mobile App 2025–2026

Personal Project

- Developed a full-stack mobile application delivering real-time spending insights and behavioral nudges at the point of purchase

- Designed a custom insight engine detecting spending anomalies by comparing daily totals against 7-day rolling averages, generating categorized alerts with severity levels
- Built RESTful API with JWT auth, service-controller separation, and PostgreSQL with Prisma ORM; deployed to Railway with automated migration on startup

Tech Stack: Node.js · PostgreSQL · Prisma ORM · JWT · Railway

BEM IT Del Portal — System Requirements Specification 2026

System Analysis & Design Course · Project Lead · Team of 4

- Led requirements elicitation and authored a complete SyRS document covering 4 integrated modules: equipment borrowing (SARPRAS), facility complaints (DEPOL), lost & found, and student aspiration management (MPM)
- Produced end-to-end documentation: Use Case Diagrams, Use Case Narratives (UC01–UC06), Sequence Diagrams, Class Diagram, DFD Level 1 & 2, Context Diagram, ERD, and Data Dictionary
- Modeled As-Is and To-Be business processes in BPMN 2.0, identifying manual workflow inefficiencies and proposing digitized alternatives with real-time status tracking and automated notifications

Tools: Draw.io · Bizagi Modeler · BPMN 2.0 · Microsoft Word

Business Process Redesign — IT Del Admissions System 2025

Business Process Management Course · Individual

- Analyzed end-to-end admission process (29 steps) using BPMN As-Is modeling, Value-Added & Waste Analysis, Fishbone Diagram, Pareto Analysis, and Queuing Analysis to identify bottlenecks contributing to a 21-day cycle time
- Designed a To-Be process model incorporating automation, task parallelism, and step elimination — improving outcomes across Time, Cost, Quality, and Flexibility dimensions

Tools: Bizagi Modeler · BPMN 2.0 · Fishbone Diagram · Pareto Analysis

REST API Testing — JSONPlaceholder API 2026

QA Portfolio Project · Individual · github.com/RotuaTampubolon/reqres-api-testing

- Designed and executed 10 structured test cases covering full CRUD operations; identified and documented 3 defects with reproduction steps, severity classification, and expected vs. actual results
- Built an end-to-end CI/CD regression pipeline using Newman CLI and GitHub Actions, enabling automated test execution on every commit and reducing manual regression effort to zero
- Produced complete QA documentation: Test Plan, Test Cases, Bug Reports, and Test Summary Report

Tools: Postman · Newman CLI · GitHub Actions · Git

Manual Testing — Login Feature Authentication 2025

QA Portfolio Project · Individual · github.com/RotuaTampubolon/saucedemo-qa

- Designed and executed 20 test cases covering positive, negative, boundary value, security, and accessibility scenarios; achieved 100% planned execution coverage
- Identified 6 defects including 2 high-severity authentication vulnerabilities; documented with full reproduction steps, severity/priority classification, and defect status tracking

Tools: Manual Testing · Test Case Design · Bug Reporting

The Kit Co. — Mountaineering Equipment Rental Platform 2025

UI/UX Designer & System Analyst · Team Project

- Designed high-fidelity prototypes in Figma for a 24/7 online mountaineering gear rental platform covering the full user journey from registration to digital checkout
- Defined the platform's operational model — physical shop as pickup point only, 100% of transactions through the web platform — and conducted requirements analysis to optimize the booking workflow

Tools: Figma · User Journey Mapping · Requirements Analysis

Student & Dormitory Management Database 2025

Database Systems Course · Individual

- Designed a normalized relational database (5+ tables) covering student biodata, dormitory assignments, and activity tracking with ERD and referential integrity constraints
- Executed and validated 15+ SQL queries for data retrieval, join operations, and reporting; generated automated summary reports reducing manual verification effort

Tools: MySQL · ERD Modeling · SQL

Student Grade Management System — C CLI Application 2025

Procedural Programming Course · Individual

- Built a C-based CLI application managing 50+ student records with file persistence, supporting grade input, classification (A–F), and edge case validation (null values, out-of-range grades, duplicate entries)

- Conducted manual functional testing across all grade logic paths; identified and resolved 3 logic errors during development

Tech Stack: C · File I/O · Modular Programming

ORGANIZATIONAL EXPERIENCE

Education Division Member

Information Systems Student Association (HIMSI) — Institut Teknologi Del | Jan 2026 – Present

- Coordinated educational programs and workshops for Information Systems students, developing planning, communication, and small-scale project leadership skills
- Collaborated with a cross-functional team to deliver learning materials, strengthening teamwork and time management abilities

CERTIFICATIONS

Software Testing Mastery 2026: QA Fundamentals to Expert — Udemy | May 2026

Credential: ude.my/UC-cca0d330-a2a8-4f2c-a88e-cce2cbf031c5